

AVAC4QGIS 1.0.0: a QGIS framework for avalanche propagation and avalanche-generated impulse-wave simulation

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Abstract

AVAC4QGIS is a QGIS framework for spatially explicit simulation of snow avalanche propagation and, optionally, the resulting impulse-wave forcing. AVAC advances a Cartesian, depth-averaged granular-flow model on a fixed DEM with Coulomb–Voellmy resistance, a curvature-dependent normal-load correction, finite terrain-tangent velocity transport, an analytical frozen-coefficient basal-source update, and static-yield arrest. Conservative shallow-layer correction-flux damping and synchronized CFL preflight complement the wet–dry finite-volume scheme. WAVE subsequently advances the standard GeoClaw shallow-water equations on a separately selected terrain/bathymetry surface. The offline coupling transfers positive avalanche volume flux and horizontal momentum across a polygon-derived shoreline. This document specifies the implemented equations, numerical treatment, geospatial preprocessing, and physical scope of the two solvers, including the assumptions and current implementation limitations that affect interpretation of their outputs.

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1 Introduction

2 AVAC4QGIS model system

AVAC4QGIS combines two depth-averaged flow models in QGIS. AVAC simulates snow avalanche release and propagation. It can run on its own or provide input to WAVE, which simulates the resulting water waves and inundation. Both solvers are based on GeoClaw, an open-source finite-volume model for geophysical flows with wetting and drying (George, 2008; Berger et al., 2011).

AVAC represents snow as a moving layer over fixed terrain. It includes basal friction and stopping, but does not model erosion, entrainment, compaction, or changes to the bed. The final snow depth represents an arrested layer only if the avalanche has stopped by the end of the simulation. In the coupled workflow, AVAC runs first and WAVE uses its saved results. Water motion does not feed back on the avalanche.

This section describes the physical assumptions, governing equations, and main numerical choices. Implementation details, including numerical stencils, file formats, and software checks, are available in the open-source repository listed in Sect. 10.

2.1 System overview and workflow

An AVAC simulation requires a digital elevation model (DEM) and one or more release polygons. The plugin transforms the polygons to the DEM coordinate reference system and uses them to create the initial snow layer. The calculation uses Cartesian coordinates, so horizontal distances must be in meters. Outputs include the evolving flow depth and velocity, their maximum values, and the final snow-depth field.

The optional WAVE calculation adds a terrain/bathymetry DEM, a lake polygon, an initial water level, and hydraulic parameters. WAVE covers the same rectangle as AVAC. Its DEM must cover this domain in the same coordinate reference system, but its elevations and resolution may differ from those used for AVAC. The lake polygon defines the initial water body and the shoreline where avalanche inflow is sampled.

The plugin prepares the shoreline source from a completed AVAC run and then starts WAVE. The saved avalanche results can be reused to examine different lake levels or hydraulic parameters. Both workflows return results to QGIS (QGIS Development Team, 2026), where each run retains its inputs, parameters, and solver identity.

Planned figure: AVAC4QGIS architecture and data flow

Show the standalone AVAC workflow, the optional WAVE calculation, their DEM and polygon inputs, the shoreline transfer, and the shared QGIS results workspace.

Figure 1: Working placeholder for AVAC4QGIS architecture and data flow.

2.2 AVAC governing equations and numerical method

2.2.1 State variables and initial snow layer

AVAC uses a Cartesian grid (x, y) over a fixed bed $B(x, y)$. The conserved state is

$$\mathbf{q}_a = (h_a, h_a u_a, h_a v_a)^\top, \quad (1)$$

where h_a is vertical flow depth and (u_a, v_a) is horizontal, depth-averaged velocity. Some avalanche models instead use thickness normal to the terrain. AVAC converts between these quantities when needed, but always stores vertical depth and horizontal momentum.

The snow is initially at rest. Within the release area, its depth can be uniform, d_0 , or corrected for elevation and slope. The elevation correction is

$$d_z = (B - z_{\text{ref}})G_z/100, \quad (2)$$

where z_{ref} is a reference elevation and G_z is the snow-depth change per 100 m. The optional slope factor is

$$f_s = \frac{\sin \theta_c - \nu \cos \theta_c}{\sin \theta - \nu \cos \theta}, \quad \theta = \tan^{-1}(\|\nabla B\|), \quad (3)$$

where θ_c and ν are prescribed release parameters. This factor applies above 25° and is zero on gentler slopes or when the denominator is numerically zero. The available depth choices are d_0 , $d_0 + d_z$, $d_0 f_s$, and $(d_0 + d_z)f_s$.

The plugin weights these depths by the fraction of each DEM cell covered by the release polygons. Coverage is estimated by subcell sampling, with holes and overlapping polygons handled together. The resulting snow volume is conservatively averaged onto the solver grid. A coarser solver spacing must be an integer multiple of the DEM spacing. If the DEM dimensions do not divide evenly, the plugin removes the minimum number of rows and columns and shares the crop as evenly as possible between opposite edges. Terrain, release, and output grids use the same retained rectangle. Invalid or negative release depths are rejected.

2.2.2 Mass and momentum equations

AVAC solves

$$\partial_t h_a + \partial_x(h_a u_a) + \partial_y(h_a v_a) = 0, \quad (4)$$

$$\partial_t(h_a u_a) + \partial_x\left(h_a u_a^2 + \frac{1}{2}gh_a^2\right) + \partial_y(h_a u_a v_a) = -gh_a \partial_x B + S_{x,r} + S_{x,g}, \quad (5)$$

$$\partial_t(h_a v_a) + \partial_x(h_a u_a v_a) + \partial_y\left(h_a v_a^2 + \frac{1}{2}gh_a^2\right) = -gh_a \partial_y B + S_{y,r} + S_{y,g}. \quad (6)$$

Here g is gravitational acceleration. The source S_r combines basal resistance with a correction for motion on steep slopes. The source S_g describes the change in horizontal velocity as the terrain tangent plane changes along the flow path.

The pressure is hydrostatic and isotropic, equivalent to an earth-pressure coefficient of one. The equations use a horizontal map grid; they are not a fully terrain-following Savage–Hutter formulation. AVAC adds slope and curvature corrections to momentum while retaining GeoClaw’s Cartesian pressure flux. These distinctions matter when comparing AVAC with models that use terrain-normal thickness and surface-following momentum.

2.2.3 Basal resistance and slope corrections

AVAC offers Coulomb, Voellmy, and cohesive Voellmy resistance (Voellmy, 1955; Salm et al., 1990; Christen et al., 2010). To express these laws on the map grid, define

$$U_h = \sqrt{u_a^2 + v_a^2}, \quad \tan \phi = \|\nabla B\|, \quad \tan \psi = -\frac{(u_a, v_a) \cdot \nabla B}{U_h}. \quad (7)$$

For moving snow, ϕ is the bed angle and ψ is the slope angle along the direction of motion, positive downhill. Terrain-normal depth is $h_n = h_a \cos \phi$, and speed along the terrain is $U = U_h / \cos \psi$. The directional bed curvature used in the source is

$$\kappa_v = \frac{u_a^2 B_{xx} + 2u_a v_a B_{xy} + v_a^2 B_{yy}}{U_h^2}. \quad (8)$$

These derivatives are computed from the fixed DEM without smoothing. Curvature corrections therefore depend on DEM resolution and elevation noise.

The basal stress is

$$\tau_b = \mu\sigma_b + I_C C + I_V \frac{\rho_a g}{\xi} U^2, \quad (9)$$

where σ_b is effective basal normal stress, ρ_a is snow density, μ is the Coulomb coefficient, ξ is the Voellmy coefficient, and C is cohesion. The indicators (I_C, I_V) are $(0, 0)$ for Coulomb, $(0, 1)$ for Voellmy, and $(1, 1)$ for cohesive Voellmy.

AVAC applies basal resistance and the steep-slope correction together as

$$\mathbf{S}_r = -h_a(a + bU_h^2) \frac{(u_a, v_a)}{U_h}, \quad U_h > 0, \quad (10)$$

with

$$\begin{aligned} a &= g \sin^2 \phi \tan \psi + \mu g \cos \phi \cos \psi + I_C \frac{C \cos \psi}{\rho_a h_a \cos \phi}, \\ b &= I_V \frac{g}{\xi h_a \cos \phi \cos \psi} + \mu \kappa_v \cos \phi \cos \psi. \end{aligned} \quad (11)$$

The first term in a corrects the excessive flow-parallel gravity produced by the Cartesian shallow-water equations on steep slopes (Hergarten and Robl, 2015). The other terms represent friction, cohesion, and velocity-dependent drag. The correction uses bed slope rather than the gradient of the evolving snow surface.

Concave curvature increases the effective normal load and thus Coulomb resistance; convex curvature reduces it (Fischer et al., 2012). The combined gravity and curvature contribution to Coulomb deceleration is

$$-\mu \cos \phi \cos \psi (g + \kappa_v U_h^2). \quad (12)$$

On convex terrain this load reaches zero at

$$U_{h,c} = \sqrt{-g/\kappa_v}. \quad (13)$$

AVAC removes the Coulomb contribution above this threshold and switches branches if the threshold is crossed during a source step. This prevents a negative normal load, a treatment also considered by Wirbel et al. (2026). Voellmy drag, cohesion, and the steep-slope correction remain active. The model does not simulate an airborne phase.

For uniform downslope motion on a plane, the gravity and resistance terms recover

$$\frac{dU}{dt} = g \left(\sin \phi - \mu \cos \phi - I_C \frac{C}{\rho_a g h_n} - I_V \frac{U^2}{\xi h_n} \right). \quad (14)$$

For non-cohesive Voellmy flow with positive net driving, $U_\infty = [\xi h_n (\sin \phi - \mu \cos \phi)]^{1/2}$. This limit provides a direct check of the source implementation.

During each basal-source step, depth, terrain geometry, material parameters, and horizontal flow direction are held fixed. The speed then satisfies

$$\frac{dU_h}{dt} = -a - bU_h^2, \quad (15)$$

which AVAC integrates analytically. The updated speed scales both momentum components equally while motion continues. Exact integration applies to this local problem; the full calculation still has errors from spatial discretization and the separate flux and source updates.

Voellmy and cohesive Voellmy set momentum to zero when the integrated speed reaches zero. A later flow update can restart motion if pressure and gravity exceed static resistance. Pure Coulomb uses a separate stopping rule: if the local source predicts zero speed on a slope that cannot support rest, it retains the incoming momentum for that source step. This fallback is not an exact continuation through zero. It also means that the stopping rules need not coincide as Voellmy drag tends to zero.

Users can assign different values of μ , ξ , and C to elevation bands. The moving source uses the bed elevation at the cell center; the normal interface yield test uses the mean adjacent bed elevation. This distinction can matter at a band boundary.

2.2.4 Velocity transport over curved terrain

A change in terrain orientation changes the horizontal components of a velocity tangent to the bed. AVAC accounts for this with a finite rotation between estimated departure and arrival tangent planes. Let $\mathbf{u} = (u_a, v_a)$ and let $\mathbf{u}^*$ be velocity after the flux update. With H_B denoting the matrix of second derivatives of bed elevation, the gradients and upward unit normals are estimated as

$$\begin{aligned} \mathbf{b}_a &= \nabla B, & \mathbf{b}_d &= \mathbf{b}_a - \Delta t H_B \mathbf{u}^*, \\ \mathbf{n}_k &= \frac{(-\mathbf{b}_k, 1)}{\sqrt{1 + \|\mathbf{b}_k\|^2}}, & k &\in \{d, a\}. \end{aligned} \quad (16)$$

The provisional velocity is lifted to the departure plane as $\mathbf{w}_d = (\mathbf{u}^*, \mathbf{b}_d \cdot \mathbf{u}^*)$. The smallest rotation that maps $\mathbf{n}_d$ to $\mathbf{n}_a$ then gives the arrival velocity. Its horizontal components define the updated momentum.

This rotation preserves the length of the lifted departure velocity and does not change snow depth. It is an identity on a planar bed or at rest, and is disabled for dry cells and the water model. In the small-step limit, its horizontal acceleration is

$$\mathbf{S}_g/h_a = -\frac{\nabla B}{1 + \|\nabla B\|^2} (\mathbf{u}^\top H_B \mathbf{u}). \quad (17)$$

The finite rotation avoids a potentially unbounded explicit integration of this acceleration. It is a local approximation of motion over the bed, not a reconstruction of every path through a grid cell. Preserving the rotated vector's length does not establish energy conservation of the complete numerical method, whose pressure flux remains Cartesian.

2.2.5 Static yield and stopping

Friction must also hold a stopped layer in place against gravity and pressure gradients. AVAC uses cell and interface yield checks inspired by the arrest treatment in D-Claw (George and Iverson, 2014). The cell-level condition for a resting layer is

$$gh_a \tan \theta \leq \mu gh_a + I_C \frac{C}{\rho_a \cos^2 \theta}. \quad (18)$$

At an interface between two wet cells, the allowable change in snow-surface elevation is

$$\Delta \eta_y = \left(\mu + I_C \frac{C}{\rho_a g \bar{h}_a \cos^2 \theta_n} \right) \Delta n, \quad \eta_a = h_a + B, \quad (19)$$

where Δn is cell spacing, $\bar{h}_a$ is mean depth, and θ_n is the bed angle across the interface. A second condition checks the full two-dimensional surface gradient in each adjacent wet cell:

$$\|\nabla_c \eta_a\| \leq \mu + I_C \frac{C}{\rho_a g h_a \cos^2 \theta_B}. \quad (20)$$

Here the subscript c denotes centered differences and θ_B is the local bed angle. This vector check prevents false arrest of diagonal flow when the separate x and y tests would each appear below yield.

Interface transport is suppressed only for states already at rest that pass both checks. The actual adjacent surface-elevation jump and local stencil estimates must also remain below the directional threshold. At a wet-dry edge, the same test uses the wet-cell depth and the height of its snow surface above the dry bed. Moving states are not frozen merely because a local predictor expects them to stop later in the step.

These conditions can maintain a non-horizontal arrested layer without smoothing its surface. They remain directional approximations on the map grid, rather than a fully rotation-invariant static-yield solution. No empirical stopping speed is imposed. Momentum is set to zero in numerically dry cells; at resolved depths, stopping follows the basal update and yield checks described above.

2.2.6 Finite-volume solution and adaptive refinement

AVAC uses GeoClaw’s augmented f -wave Riemann solver (George, 2008; Berger et al., 2011). This finite-volume method handles variable topography and wet–dry fronts, with second-order spatial corrections and transverse wave propagation. AVAC adds the static-yield checks to the normal interface solve. The plugin defaults to the Minmod limiter for Coulomb and Superbee for the Voellmy laws. The separate numerical choices used for ISeeSnow are given in Sect. 6.

Very shallow cells can develop large velocities when small momenta are divided by nearly zero depths. On nonplanar terrain, AVAC therefore reduces the higher-order correction flux smoothly as depth falls below h_c . The default threshold is 0.05 m for Coulomb and 0.10 m for the Voellmy laws; zero disables this correction. All conserved components receive the same factor, including transverse corrections. The first-order flux is unchanged. This numerical treatment uses the smooth form proposed by Kurganov and Petrova (2007), adapted here to correction fluxes and local terrain curvature. It is not an additional physical friction law.

A further flux limiter bounds outgoing transport by the depth available in the donor cell to help preserve nonnegative depth. Any residual negative depth is clipped to zero, and cells at or below $h_{\text{dry}} = 1 \times 10^{-4}$ m have zero momentum. Because clipping can add volume, mass conservation is measured explicitly.

The flux update is followed by dry-state cleanup, tangent-velocity transport, and basal resistance, in that order. These separate source updates use first-order Godunov splitting. Second-order spatial reconstruction therefore does not make the full method second-order accurate in time.

Time steps are controlled by the Courant–Friedrichs–Lewy (CFL) condition. Before accepting a step, AVAC checks wave speeds from the incoming state and reduces the step if its Courant number exceeds the configured maximum. Rejected trials do not change the solution or recorded outputs. The default target and maximum Courant numbers are 0.5 and 1.0. This check constrains incoming waves; it does not guarantee stability of every subsequent source update. Grid and time-step sensitivity must still be assessed. Real-terrain runs use outflow boundaries that allow snow to leave while blocking inflow. Analytical tests can use other boundary conditions.

AVAC supports patch-based adaptive mesh refinement (AMR) (Berger and LeVeque, 1998; Berger et al., 2011). Automatic refinement requires wet cells to exceed a speed tolerance and satisfy

$$\sqrt{h_a} U_h > \sqrt{h_v} U_{\text{tol}}, \quad (21)$$

which limits refinement caused by very thin numerical tails. Here h_v is the depth threshold used for velocity reporting, and U_{tol} is the speed tolerance for a refinement transition, set to 0.05 m s^{-1} by default. Users can also prescribe regions that require refinement. Fine grids take smaller time steps and remain synchronized with their parent grids.

Coarse-to-fine transfer reconstructs snow depth with nonnegative child values while preserving the parent mean. Fine-to-coarse transfer averages depth and both momenta over all children, including dry cells. These rules preserve avalanche volume during grid transfer, subject to the separate clipping noted above. Terrain integration on AMR grids can still introduce small geometric errors. The plugin defaults to one refinement level, as do the ISeeSnow cases reported here.

2.3 WAVE governing equations and numerical method

2.3.1 Shallow-water equations

WAVE stores water depth and horizontal momentum as $\mathbf{q}_w = (h_w, h_w u_w, h_w v_w)^\top$, where (u_w, v_w) is horizontal, depth-averaged velocity. Away from the shoreline source, it solves the following equations over terrain/bathymetry B_w :

$$\partial_t h_w + \partial_x(h_w u_w) + \partial_y(h_w v_w) = 0, \quad (22)$$

$$\partial_t(h_w u_w) + \partial_x\left(h_w u_w^2 + \frac{1}{2} g h_w^2\right) + \partial_y(h_w u_w v_w) = -g h_w \partial_x B_w + S_{x,M}, \quad (23)$$

$$\partial_t(h_w v_w) + \partial_x(h_w u_w v_w) + \partial_y\left(h_w v_w^2 + \frac{1}{2} g h_w^2\right) = -g h_w \partial_y B_w + S_{y,M}. \quad (24)$$

The source S_M represents Manning friction. The avalanche source adds water and momentum at shoreline cells, as described in Sect. 2.4. Both solvers use $g = 9.81 \text{ m s}^{-2}$.

WAVE assumes hydrostatic pressure and neglects frequency dispersion. It is intended for waves whose wavelengths are large compared with water depth. It does not resolve the three-dimensional, non-hydrostatic impact zone (Poulain et al., 2023).

2.3.2 Lake initialization and bed friction

For an initial water-surface elevation ζ_0 and lake area Ω_L , the intended resting state is

$$h_w(x, y, 0) = \begin{cases} \max(\zeta_0 - B_w(x, y), 0), & (x, y) \in \Omega_L, \\ 0, & (x, y) \notin \Omega_L, \end{cases} \quad u_w = v_w = 0. \quad (25)$$

The lake polygon is rasterized on the terrain grid. GDAL includes all cells touched by the polygon; the fallback method uses cell-center inclusion. When the WAVE grid is coarser, a cell belongs to the lake if at least half of its fine-grid mask cells are inside.

Terrain averaging near the shoreline can otherwise create an artificial wave at initialization. To reduce this effect, the plugin raises terrain samples outside the lake that lie below the waterline to $\zeta_0 + h_{\text{dry}}$. Where an interior wet cell can be retained, it also raises a one-cell inner rim. A temporary dry mask supports initialization. The rim can later be overtopped; it is not a permanent wall. Because this procedure changes shoreline geometry and initial water volume, the prepared lake must be checked before applying an avalanche source.

Bed friction is specified using Strickler coefficients $K = 1/n$, where n is the Manning coefficient. For water momentum $\mathbf{m}_w = (h_w u_w, h_w v_w)$, the local update is

$$\gamma_M = \frac{g n^2 \|\mathbf{m}_w\|}{h_w^{7/3}}, \quad \mathbf{m}_w^{n+1} = \frac{\mathbf{m}_w^n}{1 + \gamma_M \Delta t}. \quad (26)$$

It applies above the dry tolerance and below the configured friction-depth limit. The user can set different coefficients for terrain below and above the initial water level:

$$n(B_w) = \begin{cases} 1/K_w, & B_w < \zeta_0, \\ 1/K_l, & B_w \geq \zeta_0. \end{cases} \quad (27)$$

Here K_w and K_l denote water and land values.

2.3.3 Numerical solution

WAVE uses GeoClaw's standard augmented Riemann solver, second-order spatial corrections, and wetting-and-drying method (George, 2008; Berger et al., 2011). Manning friction and shoreline injection are applied

as separate source updates. The time step follows the selected CFL limits. Standard runs use zero-gradient outer boundaries; tests can select periodic or wall boundaries.

The underlying solver supports AMR, and shoreline injection is scaled by the receiving cell area. The current QGIS workflow nevertheless prepares a uniform WAVE grid. Lake initialization and shoreline forcing require their own verification, including a resting lake with negligible water-volume drift before forcing is introduced (Sect. 4.2).

2.4 One-way AVAC-to-WAVE shoreline coupling

2.4.1 Shoreline geometry and avalanche sampling

The coupling represents avalanche entry as a source in a depth-averaged water model (Vila, 1987; Ancey, 2023). The edge of the lake mask defines grid-aligned shoreline faces. Each face has length L_f and an inward normal $\mathbf{n}_f = (n_{x,f}, n_{y,f})$ pointing from land into the lake. Corner cells may receive flux through two faces. Source cells can include the initially dry stabilization rim.

This shoreline is a staircase approximation controlled by WAVE cell size. The lake polygon should leave at least one dry cell between itself and the domain boundary to avoid creating artificial boundary source faces. Shoreline resolution is therefore part of the coupling setup.

At each saved AVAC time, the plugin interpolates depth and both horizontal momenta at the shoreline-face centers. Where patches overlap, values from the finest available patch are used. Samples outside AVAC coverage are recorded and replaced by zero; preparation fails if no shoreline sample is covered. The inward volume flux per unit shoreline length is

$$q_{n,f} = (h_a u_a)_f n_{x,f} + (h_a v_a)_f n_{y,f}. \quad (28)$$

Only finite, positive inflow above the depth and flux tolerances contributes. Outward avalanche motion does not withdraw water from WAVE.

2.4.2 Volume and momentum transfer

For each face, the equivalent water-volume rate is

$$Q_{w,f} = \beta q_{n,f} L_f. \quad (29)$$

The three rates supplied to WAVE are

$$\mathbf{R}_f = (Q_{w,f}, Q_{w,f} u_{a,f}, Q_{w,f} v_{a,f})^T. \quad (30)$$

The first adds water volume; the other two add horizontal depth-integrated momentum. Multiplying the latter by water density gives physical momentum rates.

The coefficient β combines density conversion with a transfer efficiency:

$$\beta = \varepsilon_T \frac{\rho_a}{\rho_w}, \quad (31)$$

where ρ_w is water density. When $\varepsilon_T = 1$, injected mass and horizontal momentum equal the incoming avalanche values. A smaller efficiency reduces both by the same factor. The QGIS control exposes β directly as *snow-to-water damping*, with $0 \leq \beta \leq 0.4$.

This is a simplified representation of impact. Experiments show that density ratio, buoyancy, and impact velocity affect transfer (Zitti et al., 2016). The present source prescribes volume and horizontal momentum but does not separately transfer vertical momentum, energy, or a pressure impulse.

2.4.3 Time-dependent injection and conservation checks

Face contributions are summed for each receiving WAVE cell and interpolated linearly between saved AVAC times. During a WAVE step, the source is sampled at the midpoint:

$$\mathbf{q}_{w,c}^* = \mathbf{q}_{w,c}^n + \frac{\Delta t}{A_c} \mathbf{R}_c \left(t + \frac{1}{2} \Delta t \right), \quad (32)$$

where A_c is the receiving cell area. Manning friction is applied afterward.

The midpoint rule is exact within one linear interpolation interval. A step crossing an AVAC frame time can introduce an additional integration error. AVAC output frequency and WAVE time steps therefore affect the accuracy of the transferred source, as does shoreline resolution.

Each WAVE scenario records its source AVAC run, shoreline samples, receiving cells, and any missing coverage. A transfer ledger integrates the stored rates over time to report water volume and horizontal momentum. Comparing this ledger with the WAVE state helps separate errors in source transfer from errors in wave propagation (Sect. 4.3).

2.4.4 Physical limits of the coupling

The coupling is one-way. AVAC is completed before WAVE starts, and avalanche material is not removed from its saved state when it enters the lake. The method does not track submerged snow, mixing, entrainment, melting, or feedback from water to the avalanche. After injection, the transferred material behaves as water.

The model thus represents shoreline forcing followed by hydrostatic long-wave propagation. It is not a two-phase impact model. The planned laboratory comparison is needed to assess suitable transfer efficiencies and the conditions under which this approximation is useful.

Planned figure: internal shoreline transfer

Show the lake mask, inward shoreline normals, sampled avalanche flux, volume and momentum transfer, and the receiving WAVE cells.

Figure 2: Working placeholder for internal shoreline transfer.

2.5 QGIS integration, platform support, and outputs

AVAC4QGIS provides installable plugins for Windows, macOS, and Linux. Each package includes the native AVAC and WAVE solvers and their required libraries. The plugin checks and installs the appropriate runtime automatically, so routine use does not require a compiler or a separate Clawpack installation. Platform requirements and reproducibility information are provided in Appendix C.

The interface previews the snow release and initial lake. Each run stores copies of its inputs, configuration, and outputs. Users must check metric horizontal coordinates, compatible vertical datums, and adequate DEM coverage and resolution.

AVAC results include time-dependent depth, terrain-tangent speed, snow-surface elevation, and a kinetic-pressure estimate

$$p_k = \frac{1}{2}\rho_a U_{a,s}^2 / 1000 \quad (\text{kPa}), \quad (33)$$

where $U_{a,s} = \sqrt{u_a^2 + v_a^2 + (u_a B_x + v_a B_y)^2}$. This quantity is a velocity-based proxy, not a prediction of impact pressure. Summary maps show peak depth, speed, and kinetic pressure, with arrival times where available.

Temporal QGIS layers hide speed and kinetic pressure below 0.05 m. Their sharp cutoff differs from peak-speed reporting in shallow cells. The final depth map is the last saved state. WAVE outputs include water levels, wave motion, and volume and source diagnostics. Coupled runs share a QGIS timeline and retain their link to the AVAC calculation.

3 Evaluation strategy

3.1 Verification, validation, intercomparison, and demonstration

In this paper, *verification* assesses whether the numerical implementation solves the intended mathematical problem, whereas *validation* assesses whether the physical model represents observed avalanche behavior. The ISeeSnow cases are treated as an *intercomparison*: participating-model output is not assumed to be truth. A coupled application without observational data is called a *demonstration* rather than a validation.

3.2 Evidence matrix

Table 1: Evidence planned for each principal model claim.

Claim	Primary evidence	Location
AVAC solves its intended equations	Kerswell horizontal Coulomb spreading, sloping-bed Coulomb flow, and local curvature normal-load source solutions	Sect. 4.1
WAVE preserves equilibrium and moving wet/dry solutions	Lake at rest and Thacker paraboloid oscillation	Sect. 4.2
The transfer operator supplies the intended volume and momentum	Prescribed shoreline-flux ledger, closed-domain balance, grid and AMR invariance	Sect. 4.3
AVAC represents observed avalanche behavior	Laboratory or field data to be selected	Sect. 5
AVAC behaves consistently under a shared model protocol	Three official ISeeSnow cases with prescribed physical parameters; numerical choices assessed using PFT	Sect. 6
The complete workflow is fit for its stated scope	Planned coupling experiment and cross-platform reproducibility checks	Sects. 7 and 8

4 Verification

4.1 AVAC verification

The supplied manuscript illustrates AVAC verification with two finite-volume Coulomb dam-break problems derived from the exact shallow-layer solution of [Kerswell \(2005\)](#) and with closed local tests of the curvature-dependent normal-load source. The horizontal case isolates pressure-driven spreading, basal resistance, wet-dry propagation, and natural arrest. The inclined case retains the same solution structure while adding the downslope gravity source and the reduction of normal stress with slope. Taken together, the tests exercise the parts of the momentum balance that distinguish AVAC from a water-only shallow-flow solver. A third, frictionless water-limit test is reported in [Appendix A](#).

Each nominally one-dimensional problem was solved by the two-dimensional AVAC executable in a five-cell-wide strip with wall boundaries in the transverse direction. Three AMR levels were used, with two 4:1 longitudinal refinements and no transverse refinement. The horizontal case therefore progressed from a 40 mm background mesh to 2.5 mm near the front and rear boundaries; the corresponding spacings for the inclined case were 30 mm and 1.875 mm. Narrow, overlapping space-time refinement corridors followed the independently evaluated analytical boundary paths. These paths controlled mesh allocation only: neither the conserved state nor the numerical flux received analytical information. Thus the fine mesh remained concentrated near both moving boundaries while the rest of the domain stayed coarse.

For the horizontal case, the numerical front was obtained from the rightmost cell whose depth exceeded the solver dry tolerance. For the inclined case, the energy-resolved support archived by the validation notebook was used so that the reported front represents dynamically resolved material. Its post-processing threshold is $\sqrt{h_a}|U_h| > 5 \times 10^{-5} \text{ m}^{3/2} \text{ s}^{-1}$; it is independent of the solver configuration and is not a velocity ceiling. The rear boundary was the downstream edge of material whose depth remained within 0.1 % of H_0 ; this finite tolerance prevents harmless AMR interpolation perturbations from being classified as motion. Boundary errors are reported as root-mean-square errors (RMSEs) over the corresponding analytical moving interval. Depth and velocity were sampled along the strip centerline, and conservation was evaluated from the integral of flow depth per unit transverse width. The notebook for each test records the complete controls, the solver checksum, the achieved AMR level, centerline fields, and boundary time series; consequently, [Fig. 3](#) can be identified by the supplied diagnostic summary. The rendered figures and summaries are retained; regenerating any unavailable full native history requires rerunning the corresponding source and controls.

4.1.1 Kerswell horizontal Coulomb dam break

The initial condition was a static column of depth $H_0 = 1 \text{ m}$ on $-10 \text{ m} \leq x \leq 0$, with a dry bed downstream and a Coulomb coefficient $\mu = 0.1$. The upstream boundary was a wall and the downstream boundary was extrapolating. With

$$X_0 = \frac{H_0}{\mu} = 10 \text{ m}, \quad T_0 = \frac{1}{\mu} \sqrt{\frac{H_0}{g}} = 3.19 \text{ s}, \quad \tau = \frac{t}{T_0}, \quad (34)$$

the analytical front advances according to $x_f/X_0 = 2\tau - \tau^2/2$ until it arrests at $\tau = 2$ and $x_f/X_0 = 2$; the rear boundary arrests earlier, at $\tau = 1.529654$ and $x_b/X_0 = -0.721577$ ([Kerswell, 2005](#)).

AVAC reproduced both boundary trajectories and the interior depth and velocity profiles ([Fig. 3a–c](#)). During motion, the front and rear RMSEs were 0.1585 m and 0.003 11 m, respectively. At 10 s, the simulated front was at 19.831 m, compared with the analytical 20 m, and the maximum remaining speed was $1.69 \times 10^{-4} \text{ m s}^{-1}$. The depth integral was initially 10 m^2 per meter of transverse width and varied by $6.15 \times 10^{-4} \text{ m}^2$, or $6.15 \times 10^{-3} \%$. The agreement in the arrested state is especially relevant because no prescribed stopping time or fitted velocity threshold was used: arrest follows from the Coulomb and static-yield terms described in [Sect. 2.2](#).

4.1.2 Coulomb dam break on a sloping bed

The second calculation used the same released depth, but placed the column on a planar bed descending at $\theta = 5^\circ$ and increased the Coulomb coefficient to $\mu = 0.2$. In the downslope coordinate, gravity and basal resistance combine into an effective friction coefficient

$$\mu_{\text{eff}} = \mu - \tan \theta = 0.1125, \quad (35)$$

while the pressure-wave speed is based on $g \cos \theta$. Consequently, the horizontal analytical solution can be reused with $X_0 = H_0/\mu_{\text{eff}} = 8.89 \text{ m}$ and $T_0 = \sqrt{H_0/(g \cos \theta)}/\mu_{\text{eff}} = 2.84 \text{ s}$. This equivalence provides a direct test of the balance between the topographic source and Coulomb resistance implemented in AVAC, rather than merely a second wet-dry propagation test.

The resulting dimensionless trajectories and profiles closely followed the same analytical curves as the horizontal case (Fig. 3d–f). The moving-front and rear RMSEs were 0.1232 m and 0.0342 m, respectively. At 6 s, the illustrated run placed the resolved front at 17.6253 m, compared with the analytical 17.776 m; the rear positions were -6.5209 m and -6.4082 m . The maximum resolved speed had decreased to $1.14 \times 10^{-3} \text{ m s}^{-1}$. Over the calculation, the 9.99 m^2 initial depth integral varied by $9.07 \times 10^{-4} \text{ m}^2$, corresponding to $9.08 \times 10^{-3} \%$. These are the supplied historical diagnostic definitions, not a recomputation using the current release's controls.

The two Coulomb tests are complementary. The horizontal calculation shows that AVAC propagates and arrests a finite granular mass without gravitational driving along the bed, whereas the inclined calculation shows that the same behavior is recovered after the driving and resisting accelerations are combined. Their common dimensionless solution therefore links the observed agreement to the implementation of the AVAC source terms, rather than to a case-specific parameter adjustment.

4.1.3 Local curvature normal-load source

The curvature-dependent normal-load term was checked independently of the finite-volume flux by compiling the same Fortran rheology module used by AVAC and comparing its scalar frozen-coefficient update with closed solutions. A material element with initial map speed 8 m s^{-1} and $\mu = 0.30$ was evaluated for 4 s at the nadir of planar, concave-circular, and convex-circular tracks. The circular radius was 40 m, giving $\kappa_v = \pm 0.025 \text{ m}^{-1}$, and the analytical equation was

$$\frac{dU_h}{dt} = -\mu \left(g + \kappa_v U_h^2 \right). \quad (36)$$

The maximum absolute speed errors were below $3.2 \times 10^{-15} \text{ m s}^{-1}$. Repeating a 2 s update as eight equal source substeps changed the result only at machine precision.

A zero-force test of the isolated *scalar basal routine* leaves its input map speed unchanged. It must not be interpreted as the behavior of the full current source step, which first performs the separate terrain-tangent transport. The production regression tests additionally exercise affine-bed identity, tangent-speed preservation, the differential connection limit, rotated two-dimensional geometry, physical boundaries and patch partitioning. These checks test distinct operators rather than replace a resolved curved-DEM trajectory or establish convergence of their split composition.

The circular-track values above verify the normal-load term for frozen local geometry. Their convex state remains below the loss-of-contact threshold in Eq. (13); that particular example does not exercise the branch switch. Compiled source-stopping tests cover the corrected Voellmy and cohesive-Voellmy zero-result handling, including near-dry states and unchanged Coulomb/water controls. The released test sources retain these separate scopes explicitly.

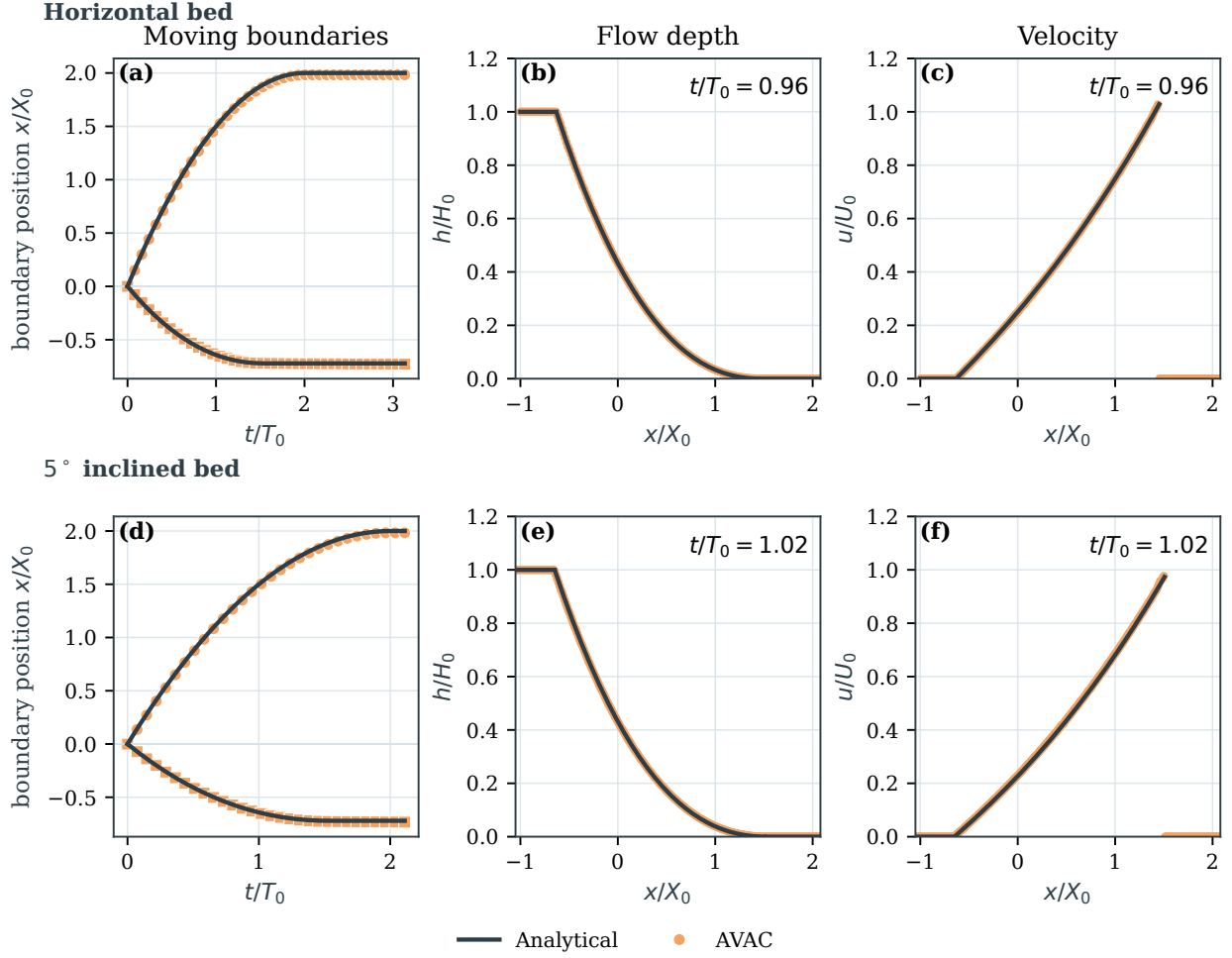


Figure 3: AVAC verification against the Coulomb dam-break solution of [Kerswell \(2005\)](#). The upper row shows the horizontal-bed case and the lower row the 5° inclined-bed case. Panels (a,d) compare the front (circles) and rear (squares) trajectories; panels (b,e) compare flow depth; and panels (c,f) compare downslope velocity at $t/T_0 \simeq 1$. Dark lines denote the analytical solution and orange symbols denote AVAC centerline results. AMR provides finest longitudinal spacings of 2.5 mm and 1.875 mm in the two rows, respectively. These are the supplied pre-1.0.0 verification fields (executable prefix 6e85bee646c3), not a new release-wide AMR run.

4.2 WAVE verification

4.2.1 Lake at rest

The first WAVE test asks whether the lake initialization and the discretized bathymetric source remain balanced before any avalanche source is applied. The bed contains the smooth bump used by the Baines steady-flow test, while the initial free surface is horizontal at 2 m. The 25 m by 1 m domain is resolved with 0.05 m cells, the longitudinal boundaries are closed, and the transverse boundaries are periodic. The coupling file contains no active source. Consequently, the exact solution is a lake at rest for the complete 10 s calculation.

The free-surface errors rose during the initial numerical adjustment and then remained nearly constant (Fig. 4a). The largest L_∞ and L_2 errors over all saved times were 6.37×10^{-4} m and 6.11×10^{-5} m, respectively. Thus the topographic balance introduces a submillimetric disturbance, but no growing wave. This test is especially relevant to the coupled workflow: it places an upper bound on the background signal against which an avalanche-generated impulse must be distinguished.

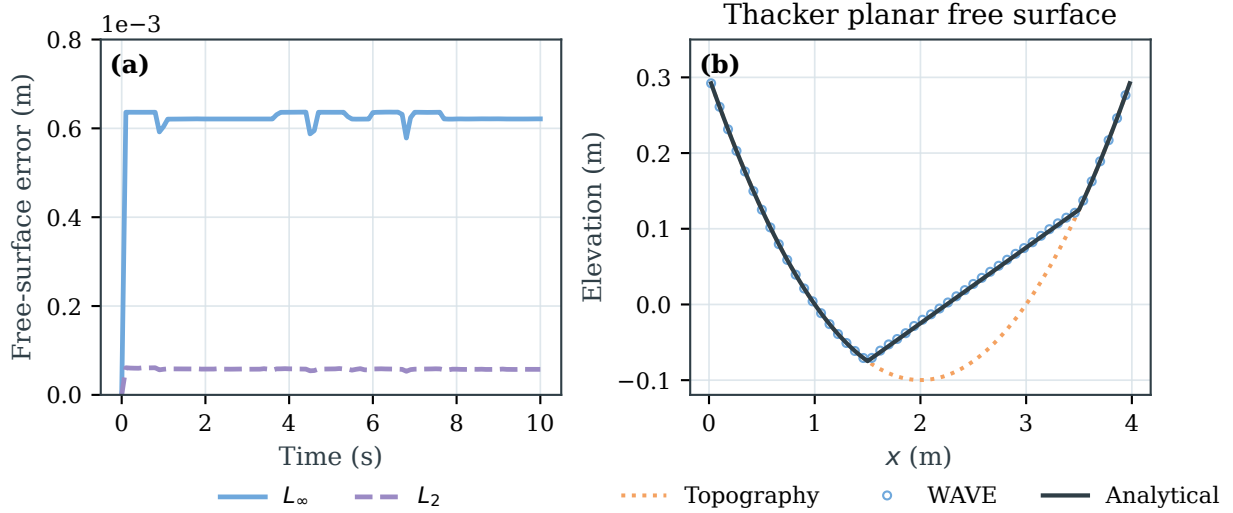


Figure 4: WAVE verification. (a) L_∞ and L_2 free-surface errors for the lake-at-rest test over a smooth bump. (b) Centerline free surface after the third period of the planar Thacker oscillation in a parabolic basin. Black is the analytical solution, blue open circles are WAVE, and orange dotted line is the topography. Legends are placed outside the plotting regions.

4.2.2 Thacker planar oscillation in a paraboloid

The still-water calculation does not exercise shoreline motion. We therefore use the exact planar free-surface oscillation in a parabolic basin derived by [Thacker \(1981\)](#), with reference values generated independently through the SWASHES collection ([Delestre et al., 2013](#)). A 4 m square basin is represented by a 100×100 grid, giving 0.04 m cells, and wall conditions are imposed on all four sides. The calculation covers three complete oscillation periods, so that phase, amplitude, and repeated wetting and drying all contribute to the final error.

Figure 4b compares the WAVE and analytical centerline free surfaces over the parabolic topography after the third period. The WAVE centerline depth follows the analytical wet support and profile closely; the depth RMSE is 1.27×10^{-3} m, and the maximum absolute error is 2.84×10^{-3} m. Together with the lake-at-rest result, this shows that WAVE both maintains a nominally stationary lake and transports a moving shoreline through a genuinely two-dimensional basin without a systematic phase shift visible at the reported resolution.

The remaining analytical profiles and numerical diagnostics are collected in Appendix B. They broaden the evidence to steady hydraulic transitions, shocks, friction, dry fronts, variable channel width, AMR accuracy, and parallel reproducibility without interrupting the stationary-to-moving-lake progression used here.

4.3 Coupling verification

The component tests above verify the two solvers separately, but not the transfer between them. We therefore prescribed a spatially uniform synthetic AVAC state along a straight shoreline of length $L = 0.8$ m. Over $T = 1$ s, its depth was $h_a = 0.1$ m and its velocity was

$$(u_a, v_a) = A(t)(0.02, 0.01) \text{ m s}^{-1}, \quad A(t) = \sin(\pi t/T) \left(1 + \frac{t}{4T}\right). \quad (37)$$

The pulse is zero at both ends and has closed-form volume and momentum integrals. It was written as native AVAC frames, sampled by the production shoreline converter, serialized in the conservative source format, read by the WAVE Fortran source module, and applied by the current WAVE executable. Thus the test exercises the complete numerical transfer rather than only the Python rate calculation.

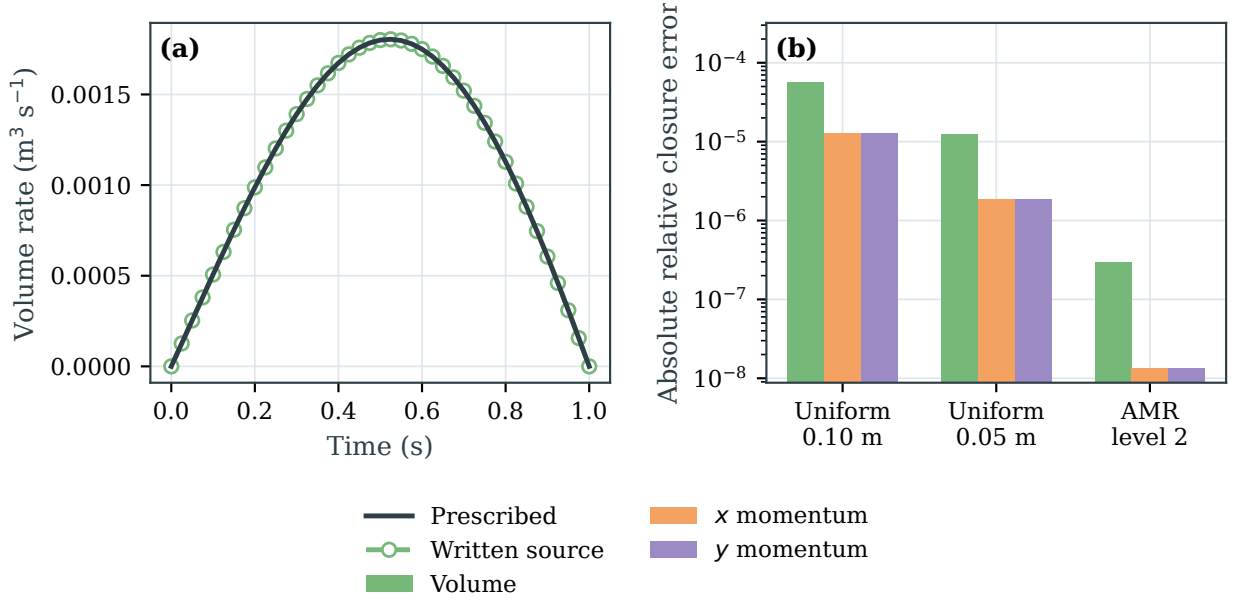


Figure 5: Numerical verification of the AVAC-to-WAVE coupling. (a) Prescribed and serialized volume-rate histories. (b) Absolute relative closure errors for volume and both depth-momentum components on two uniform grids and a level-2 AMR grid. The complete path includes native AVAC frames, the production converter and source file, the Fortran source update, and native WAVE control volumes.

For a uniform active shoreline, the expected transferred volume, depth-integrated momentum, and physical water momentum are

$$V_w = \beta L \int_0^T h_a u_n dt, \quad (38)$$

$$I_x = \beta L \int_0^T h_a u_n u_a dt, \quad P_x = \rho_w I_x, \quad (39)$$

$$I_y = \beta L \int_0^T h_a u_n v_a dt, \quad P_y = \rho_w I_y, \quad (40)$$

where u_n is the velocity component directed into the lake and $\beta = 1$ for this verification. Equation (37) gives $V_w = 1.14592 \times 10^{-3} \text{ m}^3$, $I_x = 2.02827 \times 10^{-5} \text{ m}^4 \text{ s}^{-1}$, and $I_y = 1.01413 \times 10^{-5} \text{ m}^4 \text{ s}^{-1}$. The WAVE domain was a frictionless, periodic 2 m by 1 m basin with an initial uniform depth of 0.5 m. Periodicity removes boundary losses, so changes in the three globally integrated conserved variables can be compared directly with the written source ledger.

The prescribed and serialized volume-rate histories are visually coincident (Fig. 5a). WAVE was run on uniform 0.10 m and 0.05 m grids and with two AMR levels from a 0.10 m base grid to a 0.01 m finest grid. Integration of the non-overlapping native control volumes gave a maximum absolute relative closure error of 5.61×10^{-5} across volume and both momentum components (Fig. 5b). The AMR run reached level 2 and reduced the largest closure error to 2.95×10^{-7} ; the result therefore confirms that the format-2 total rates remain conservative when the receiving cell area changes.

These tests verify numerical transfer and conservation, but they do not validate the physical assumption that an entering avalanche can be replaced by an equivalent one-way water source. That question requires the planned laboratory experiment in Sect. 7; the distinction between numerical coupling verification and physical coupling validation is maintained throughout this paper.

5 AVAC validation against observations

Before drafting this section, we must identify a comparable avalanche-model validation paper with accessible release geometry, terrain, material parameters, and observations of runout, front motion, velocity, or deposit. We should prefer an open dataset and a case that permits either independently prescribed parameters or calibration on one event followed by validation on another. A single case fitted and assessed with the same outputs must be described as calibration or evaluation, not independent validation.

6 ISeeSnow avalanche-model intercomparison

6.1 Protocol and common metrics

The ISeeSnow v1.0 exercise provides common idealized and real topographies, release areas, rheological parameters, and submitted peak fields from several depth-integrated avalanche models (Wirbel et al., 2026). We ran AVAC on the supplied 5 m grid without retuning the prescribed physical parameters. Numerical limiter and shallow-flow controls were assessed using peer PFT comparisons, as distinguished below. The three calculations are the idealized Voellmy case ($\mu = 0.4$, $\xi = 2000 \text{ m s}^{-2}$), the real-topography Voellmy case ($\mu = 0.2$ with the same ξ), and the idealized Coulomb-only case ($\mu = 0.4$ and no turbulent resistance). In all three cases, the prescribed terrain-normal release thickness is 1.5 m.

The two idealized cases share the same inclined plane, variable-width ravine, release area, and flat foreland, while the real-terrain case uses a distinct mountain catchment and release polygon (Fig. 6). Thus the first and third calculations contrast Voellmy and Coulomb-only resistance on a common geometry, using the model-specific numerical controls stated below, whereas the first and second retain Voellmy resistance while changing both terrain and the prescribed Coulomb coefficient.

Because ISeeSnow prescribes release thickness normal to the terrain whereas AVAC advances vertical map-cell depth, initialization applies the inverse geometric conversion

$$h_a(x, y, 0) = \frac{h_n(x, y, 0)}{\cos \phi(x, y)}. \quad (41)$$

The prescribed normal release volume is therefore represented consistently on sloping cells before the calculation begins.

AVAC stores vertical depth and horizontal map-coordinate momentum, whereas ISeeSnow prescribes terrain-normal peak flow thickness (PFT). We therefore convert vertical peak depth back to terrain-normal PFT cell by cell as $\text{PFT}_n = \text{PFT}_h \cos \phi$. PFV is the physical terrain-tangent magnitude consistent with Eq. (7), $[u_a^2 + v_a^2 + (u_a \partial_x B + v_a \partial_y B)^2]^{1/2}$, evaluated at every solver step before taking the temporal maximum. The release state at $t = 0$ is included explicitly in the PFT maximum, and subsequent maxima are obtained from the native peak-output calculation. PFV is zero for vertical depth at or below 0.05 m, uses the smooth diagnostic transition described in Sect. 2.2 between 0.05 m and 0.20 m, and is the exact terrain-tangent velocity at larger depth. This follows the model-specific minimum-depth criteria documented by ISeeSnow. The AVAC run configuration sets the shared GeoClaw speed limit to 10^{99} m s^{-1} , so neither the evolving state nor the reported PFV is clipped by a finite velocity ceiling.

Two peer populations serve different questions. The scalar overview uses the published Table C1 core group: 11 idealized-Voellmy, 10 real-terrain, and 11 Coulomb submissions. It compares thalweg runout, maximum PFT and maximum PFV. Runout is the furthest thalweg coordinate with normal PFT exceeding 0.5 m; it is not the area of a 0.01 m support mask. The off-scale Coulomb PFV of r.avaflow is retained in all scalar statistics and annotated in the figure, not discarded.

Spatial PFT comparison, the primary numerical-selection criterion, instead uses only rasters with exactly matching dimensions, cell size and sample coordinates: 9, 8 and 8 peers for the three cases. No field is shifted or resampled to improve agreement. Its dimensionless score is

$$E_{\text{PFT}} = \text{median}_i \left[\frac{\text{RMSE}(\text{PFT}_{\text{AVAC}}, \text{PFT}_i)}{\text{median}_{j \neq i} \text{RMSE}(\text{PFT}_i, \text{PFT}_j)} \right], \quad (42)$$

using full-grid errors and the recorded common nodata treatment. It is not the unnormalized median RMSE in meters, nor an observational accuracy score. The broader scalar population must not be confused with this spatially aligned population.

All cases use a uniform 5 m grid, second-order spatial updating, a 1200 s horizon and 10 s saved-state intervals. The two Voellmy cases use Van Leer and target CFL 0.50; Coulomb uses Minmod and target CFL 0.25. The correction-flux depth controls are respectively 0.10 m and 0.05 m; the solver dry tolerance is $1 \times 10^{-4} \text{ m}$ and the rejection ceiling is CFL 1.0.

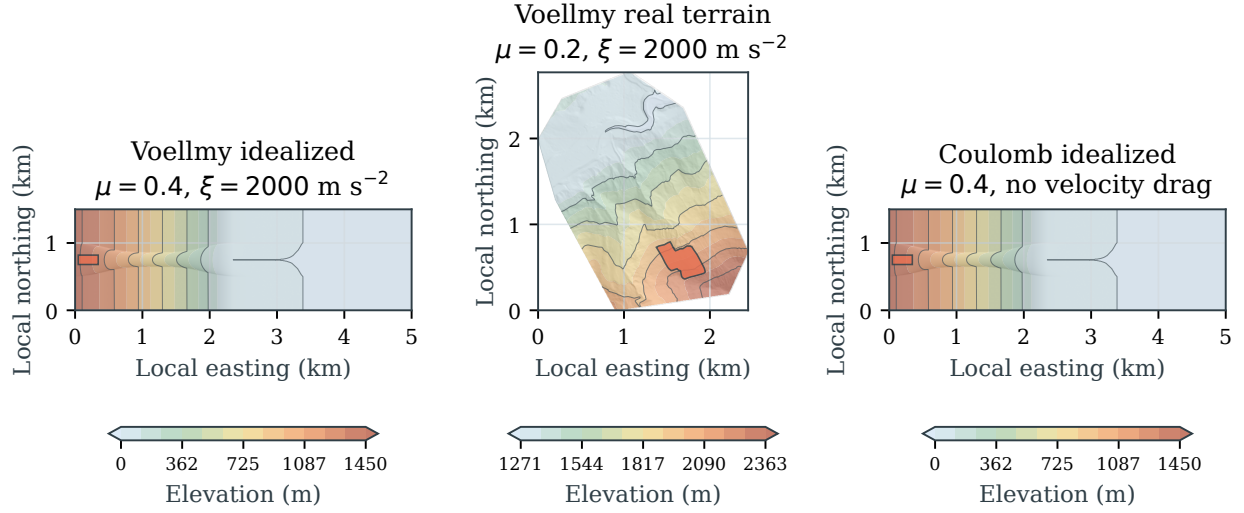


Figure 6: The three AVAC ISeeSnow configurations, drawn from the supplied benchmark DEMs and release polygons. (a) Idealized topography with Voellmy resistance, (b) real topography with Voellmy resistance, and (c) the same idealized topography with Coulomb resistance only. Red polygons denote the common 1.5 m terrain-normal release thickness. The maps are an original rendering of the inputs used in Wirbel et al. (2026).

6.2 Current fields and comparison with peers

Table 2: Accepted ISeeSnow results carried into version 1.0.0. The PFT score uses aligned spatial peers; the figure uses Table C1 scalars.

Case	Runout (km)	Peak PFT (m)	Peak PFV (m/s)	E_{PFT}
Idealized Voellmy	2.31005	8.53216	39.6255	0.865517
Real-terrain Voellmy	2.09712	21.03670	67.6648	0.907803
Idealized Coulomb	3.15004	7.21983	110.9650	0.911514

For idealized Voellmy, runout and peak PFT lie within the scalar peer interquartile intervals, 2.305 km to 2.335 km and 7.635 m to 9.285 m. Peak PFV is slightly above the upper quartile of 39.285 m s^{-1} , but below the peer maximum of 41.55 m s^{-1} . For real terrain, runout and peak PFT are also within their interquartile intervals. PFV lies above the upper quartile of 52.465 m s^{-1} , within the full 40.87 m s^{-1} to 80.23 m s^{-1} range.

The Coulomb runout lies within the peer interquartile interval 3.134 34 km to 3.226 52 km. Peak PFT is slightly above its upper quartile of 7.125 m, while remaining within the 3.95 m to 9.97 m range. PFV is likewise just above its upper quartile of $110.7155 \text{ m s}^{-1}$. Thus the accepted Coulomb runout is comparable with the central peer population, but the deposit crest is not identical to a peer median. The peak normal thickness is 7.219 83 m; its cell retains 7.189 42 m at the horizon with zero momentum. The recorded direct-interface guard correction did not alter this Coulomb trajectory relative to the preceding experimental V1 cohort: the complete saved native histories and final PFT/PFV fields match. The crest is not attributed to that separately demonstrated guard defect.

The peer products are maxima over time, not observed final-deposit histories. Different peer crests occur at different coordinates, so a pointwise median can be lower than the median of their individual maxima. Neither a peer envelope nor the scalar agreement in Table 2 identifies a unique physically correct deposit.

6.3 Conservation, practical rest and remaining sensitivity

The native relative volume changes $(V_f - V_0)/V_0$ are 1.6463×10^{-9} , 6.0960×10^{-10} and -6.4338×10^{-10} for idealized Voellmy, real-terrain Voellmy and Coulomb. Practical-rest confirmation occurs at 590 s, 520 s and 100 s, respectively, and remains satisfied at subsequent saved checks through the horizon. This criterion requires less than 1% of the release volume to be moving faster than 0.01 m s^{-1} in the resolved depth band above 0.05 m, for at least three saved times. It is not zero velocity in every wet cell. Final resolved moving volumes are approximately 70.09 m^3 , 231.35 m^3 and zero; motion below the reporting-depth threshold is audited separately.

The correction that honors Voellmy's exact zero source result removed the reproduced extremely large near-dry post-source states. Both full Voellmy observation-only audits reproduced all 484 saved native artifacts and the PFT/PFV fields; no invalid accepted post-source states were observed. However, real terrain still has a raw all-step terrain-tangent maximum of 90.97 m s^{-1} in the 0.05 m to 0.10 m depth band, distinct from its filtered PFV of 67.66 m s^{-1} . Neither saved frames nor filtered peak fields bound every intermediate split state.

A same-source Voellmy target-CFL refinement from 0.50 to 0.25 reduced the real-terrain thin-layer pulse and improved its normalized PFT score by about 1.52%, but worsened the idealized PFT score by about 1.84%. PFV and local fields changed non-uniformly. Accordingly, the existing controls are retained; two timestep levels do not establish a convergence rate or universal stability. The improved PFT comparisons, the separate flat-control evidence and these remaining sensitivities are reported together rather than treating peer agreement as a substitute for numerical verification.

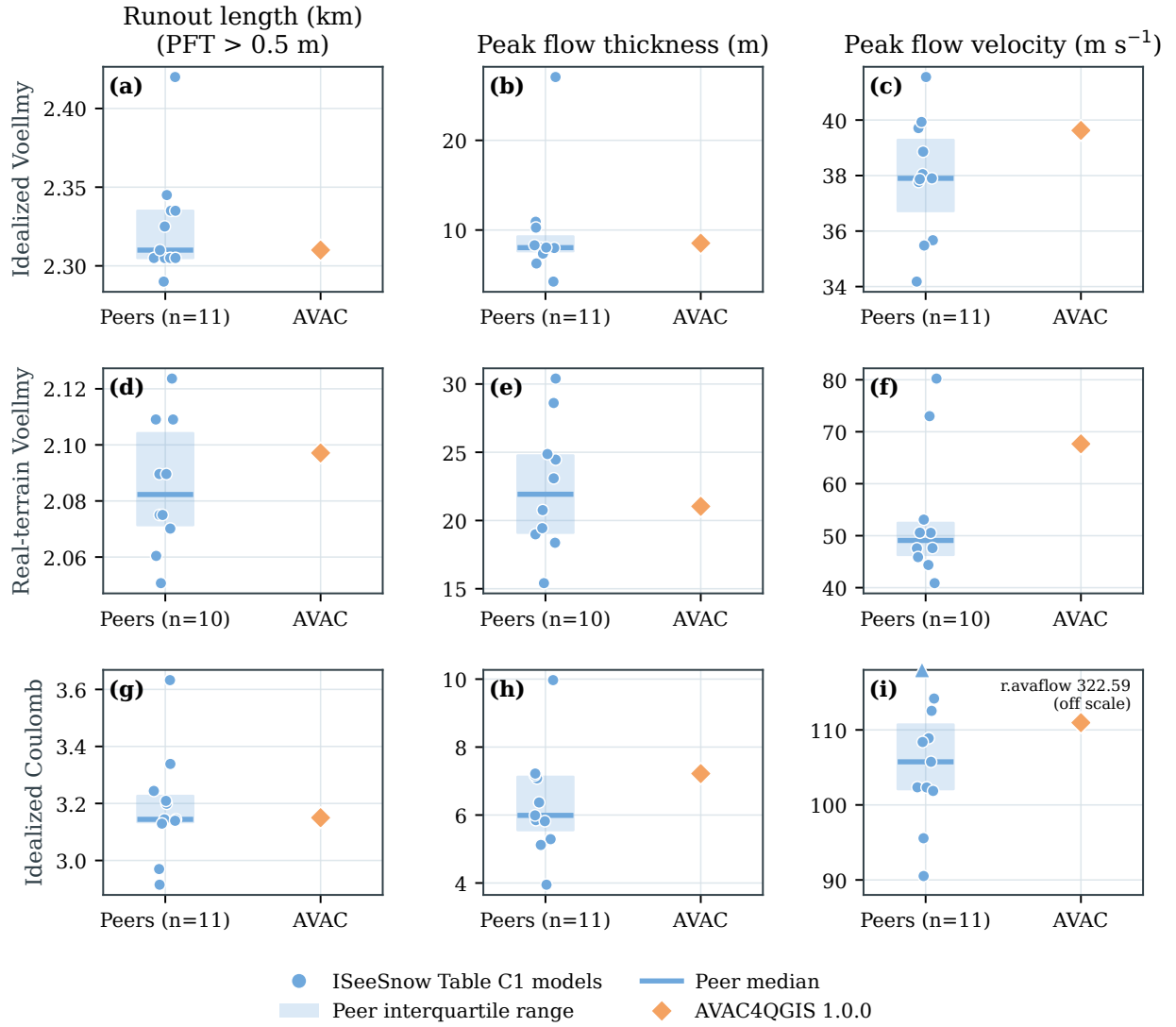


Figure 7: Version 1.0.0 in the ISeeSnow Table C1 core-model ensemble. Rows show idealized Voellmy, real-terrain Voellmy and idealized Coulomb; columns show thalweg runout for PFT greater than 0.5 m, peak PFT and peak PFV. Blue points, bands and lines are peer scalars, interquartile ranges and medians; orange diamonds are the accepted AVAC fields. All 11/10/11 peer values are retained. The off-scale Coulomb velocity is annotated and included in the statistics. This is a replot of authenticated completed-run metrics, not a newly executed validation experiment.

7 Coupled avalanche–impulse-wave validation experiment

Write after the planned laboratory experiment is completed. Planned subsections:

1. experimental geometry, material properties, release, water depth, and measurement uncertainty;
2. AVAC motion and shoreline-arrival state;
3. transferred volume and momentum ledger;
4. WAVE gauge histories, amplitudes, wavelengths, and runup;
5. sensitivity to β , grid spacing, shoreline representation, and AVAC output interval;
6. discrepancies attributable to hydrostatic and one-way coupling assumptions.

Reserve the terms “coupling validation” and “validated coupled model” until the experiment and acceptance metrics justify them.

8 Discussion

Draft once all evidence is available. Planned subsections:

1. what component verification, AVAC validation, ISeeSnow, and the coupling experiment establish;
2. physical meaning and limitations of the one-way shoreline transfer;
3. hydrostatic/non-dispersive limitations of WAVE;
4. grid, shoreline, and temporal-sampling sensitivities;
5. accessibility and reproducibility through QGIS and managed runtimes;
6. future two-way, multilayer, or dispersive developments.

Scientific equivalence across Windows, macOS, and Linux should be assessed with a common test case and declared numerical tolerances.

9 Conclusions

Four short paragraphs: the model-system contribution; evidence for AVAC and WAVE; evidence from ISeeSnow and the coupling experiment; and the final scope-of-use statement.

10 Code and data availability

The source code and plugins for Windows, macOS, and Linux are available as version 1.0.0 at <https://github.com/cgotelli/AVAC-QGIS/releases/tag/v1.0.0>; the integrated development branch is `main`. The editable manuscript, bibliography, figures and replot script are stored under `docs/article`. The release also provides the compiled methods PDF, platform-specific plugin ZIP files, SHA-256 checksums, and runtime manifests. The main source tree retains the validation drivers and tests; third-party components and their supplied licenses are described in `THIRD_PARTY_NOTICES.md`.

The ISeeSnow input geometry and peer submissions originate from the common protocol of Wirbel et al. (2026). The updated scalar figure retains the archived input-data hash, exact AVAC solver identity, Table C1 values and portable replot data. It can be redrawn without a solver run. Selected final PFT/PFV fields, controls, diagnostic reports and earlier worktree patches were preserved in compact local evidence archives during release preparation; bulk intermediate native histories were removed. Historical audit reports remain records of the completed checks, but repeating their full byte-level audits requires regenerating those histories. This repository deposit is not claimed to have a DOI or to be a complete permanent archive of every intermediate field. Planned observational and laboratory-validation data are not represented as available results.

A Additional AVAC verification

As an end-member test, AVAC was also configured with zero basal friction and no avalanche rheology and compared with the exact finite-volume dam-break solution of [Ancy et al. \(2008\)](#). A triangular prism of water with $H_0 = 1$ m was released on a frictionless 10° planar bed; its initial depth was $h(x, 0) = H_0(1 - x/x_b)$ for $x_b \leq x \leq 0$, where $x_b = -H_0/\tan(10^\circ)$. The calculation used the same three-level, front-focused AMR procedure as the main AVAC tests on a domain extending from $x = -10$ m to $x = 60$ m, progressing from a 40 mm background spacing to 2.5 mm near the head and tail. This case is placed in the appendix because it verifies the frictionless shallow-water limit shared with GeoClaw, rather than avalanche-specific resistance.

A constant 10 m vertical datum was retained in the planar-bed input used for this benchmark. The present AVAC configuration sets GeoClaw’s reference sea level to -10^4 m, safely below the computational terrain, so the translation is no longer required to keep newly refined dry cells dry. It is nevertheless immaterial to the comparison because a uniform vertical translation changes neither bed gradients, water depth, nor the analytical solution.

The extended domain retained both moving boundaries throughout the complete 5 s comparison. Over that interval, the front and rear RMSEs were 0.062 66 m and 0.080 56 m, respectively. The maximum initial-depth discrepancy on the uniform diagnostic grid was 3.31 mm. The initial depth integral, $2.835\,64\text{ m}^2$ per meter of transverse width, differed from the exact value by $1.31 \times 10^{-7}\text{ m}^2$ and varied by $4.15 \times 10^{-4}\text{ m}^2$ over the full simulation, or 0.0146 %. These errors include the interpolation from the front-focused AMR hierarchy to the one-row diagnostic grid.

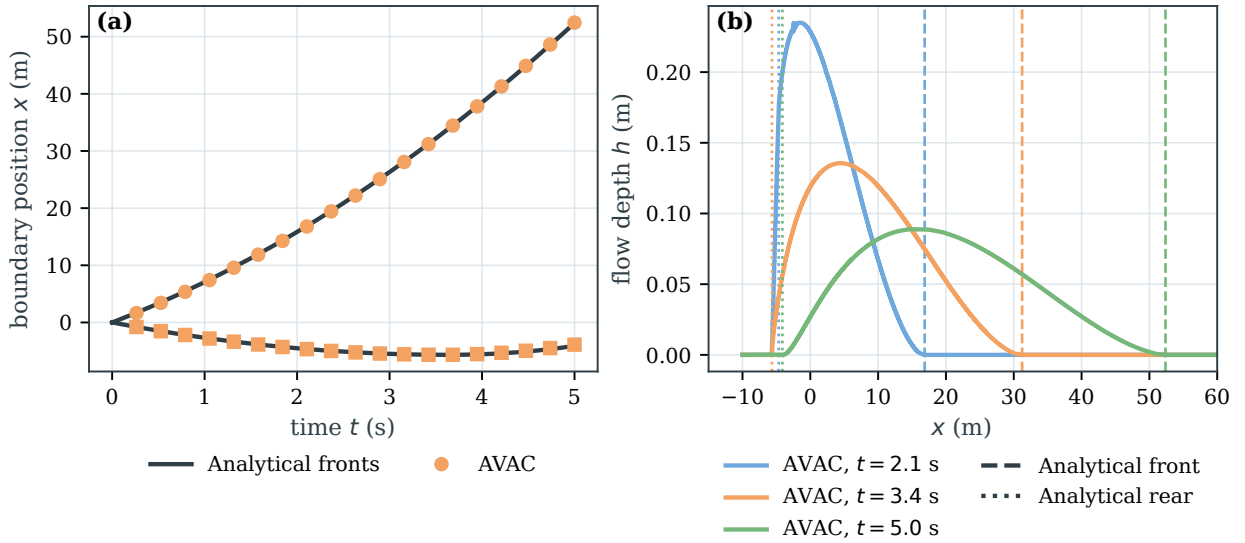


Figure 8: Frictionless AVAC water-limit verification against [Ancy et al. \(2008\)](#). (a) Analytical and numerical positions of the advancing front and rear boundary over the complete 5 s simulation. (b) AVAC depth profiles at approximately 2.1 s, 3.4 s, and 5 s; dashed and dotted vertical lines mark the corresponding analytical front and rear positions. The profiles are produced from saved centerline fields and are not solver inputs. This is the supplied historical checkpoint, not a newly executed version 1.0.0 benchmark.

The water-limit result completes a progression from frictionless acceleration to Coulomb spreading on horizontal and inclined beds. It confirms that the underlying wet-dry and topographic terms retain their expected shallow-water behavior before avalanche-specific resistance is activated. Full numerical controls and machine-readable error summaries accompany the executable notebook, while the figure-generation notebook reads only those archived products.

B Additional WAVE verification

The two main-paper tests were selected because they most directly represent a stationary and an oscillating lake. The remaining executable notebooks cover steady hydraulic transitions, shocks, friction, a dry dam break, variable channel width, and the accelerating sloping-bed solution. Table 3 collects their principal depth errors using each benchmark’s natural one-dimensional or cross-section-mean diagnostic. The SWASHES solutions were generated from the pinned analytical implementation (Delestre et al., 2013); no numerical reference solver was used.

Figure 9 shows selected saved numerical profiles beside their analytical references. The panels cover one-dimensional hydraulic transitions, wet–dry propagation, and the full error histories for the steady Baines bump and lake-at-rest configurations.

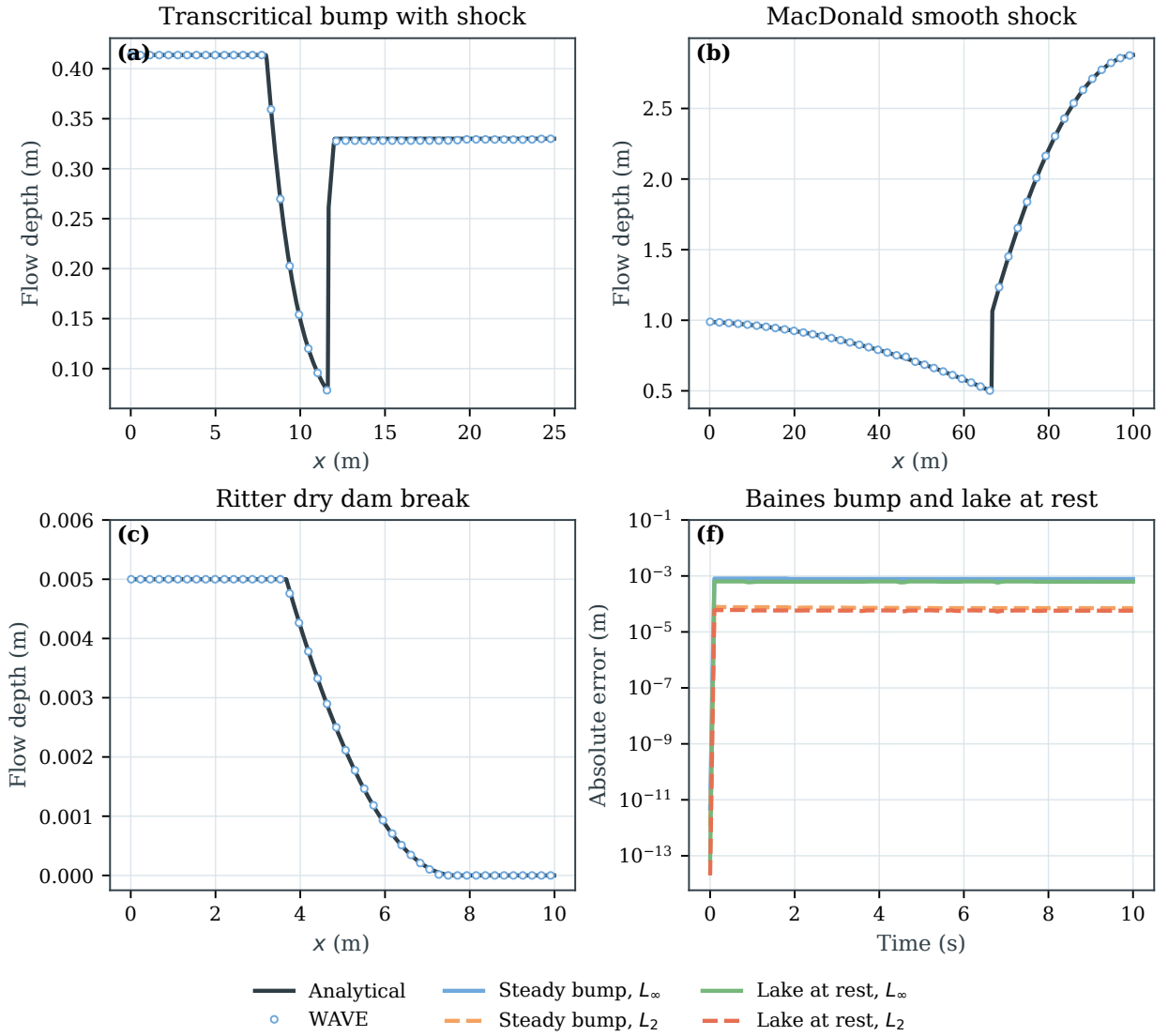


Figure 9: Additional WAVE analytical verification. (a) Transcritical flow over a bump with a shock, (b) the MacDonald smooth-transition and shock solution, (c) the Ritter dry-bed dam break, and (f) the Baines steady-bump and lake-at-rest error histories. Dark curves are analytical solutions, blue open circles are WAVE, and the pastel curves in (f) distinguish the two error norms and stationary tests.

WAVE was also applied to the frictionless dry-bottom sloping-bed solution of Ancy et al. (2008) with 0.005 m cells. Before its advancing front reached the tutorial’s open boundary, the front and rear RMSEs were 0.170 m and 0.0274 m, respectively, while the initial depth differed from the exact profile by at most 5.63×10^{-13} m. This independently exercises the water executable on the same accelerating wet–dry problem

Table 3: Additional WAVE analytical-verification results. “Centerline” applies to the two-dimensional Thacker basin, and “section mean” to the variable-width pseudo-2D channels.

Case and metric	Δx (m)	RMSE (m)	Maximum error (m)
Transcritical bump with shock, depth	0.05	4.26×10^{-3}	9.15×10^{-2}
MacDonald smooth shock, depth	0.20	7.72×10^{-3}	1.68×10^{-1}
Ritter dry dam break, depth	0.02	1.07×10^{-5}	7.15×10^{-5}
Thacker paraboloid, centerline depth	0.04	1.27×10^{-3}	2.84×10^{-3}
Pseudo-2D supercritical, section mean	0.10	3.30×10^{-2}	1.05×10^{-1}
Pseudo-2D subcritical, section mean	0.20	5.18×10^{-2}	8.36×10^{-2}
Baines steady bump, depth	0.05	7.07×10^{-5}	7.80×10^{-4}
Lake at rest, free surface	0.05	5.75×10^{-5}	6.21×10^{-4}

used for AVAC in Appendix A.

Finally, a flat-bed dry dam break compared a 0.05 m uniform grid, a 0.005 m uniform reference, and level-2 AMR with the same finest spacing. AMR reduced the RMS depth difference from 1.90×10^{-3} m to 2.30×10^{-4} m. Repeating both the fine uniform and AMR calculations with one and eight OpenMP threads produced byte-identical final native and fixed-grid files. Native non-overlapping AMR integration, rather than a coarse visualization grid, gave a maximum relative mass change of 4.68×10^{-7} across the uniform and AMR configurations.

These last two diagnostics are summarized in Fig. 10. The sloping-bed calculation tests an accelerating wet–dry boundary before the open boundary is reached, while the dam-break comparison shows the improvement obtained by concentrating the fine spacing near the front. Thread-count equality is reported separately because coincident curves alone cannot establish bitwise reproducibility.

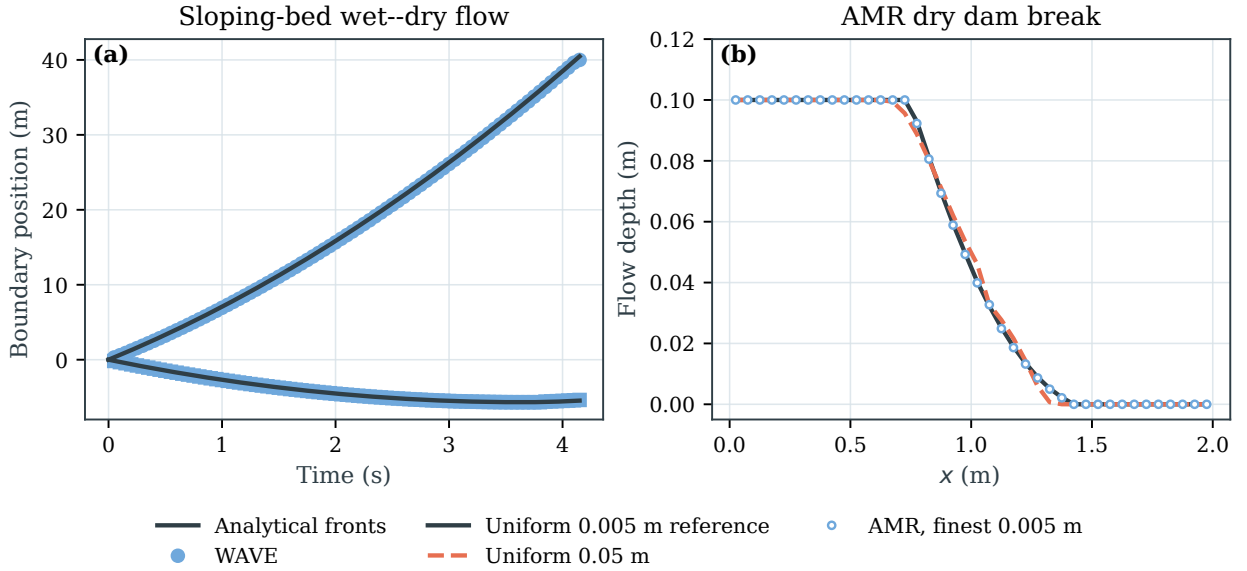


Figure 10: WAVE wet–dry and numerical diagnostics. (a) Front and rear trajectories for the frictionless sloping-bed solution before the advancing front reaches the open boundary. (b) Final dry-dam-break depth on the uniform coarse grid and level-2 AMR grid, compared with the uniform fine reference. The AMR markers use the same 0.005 m finest spacing as the reference.

C Cross-platform reproducibility

Version 1.0.0 provides plugin packages for Windows, macOS, and Linux. Each package includes native solvers, Clawpack 5.14.0, required libraries, and component licenses. Supported architectures, operating-system requirements, checksums, and runtime inventories are documented with the release.

The Windows AMD64 build supports QGIS 3.40 or newer. Its earlier isolated functional test used QGIS 3.40.11-Bratislava: AVAC and WAVE runtimes installed automatically and an AVAC run completed. That offscreen QGIS process subsequently crashed at shutdown, so its functional pass is not a clean-exit certification. The version 1.0.0 packaging and isolated-runtime installation checks are separate from that earlier GUI run.

For the Windows package, the AVAC and WAVE executable SHA-256 values are, respectively,

```
55a8c4baf0cc74d9a8ca4c582f361eb85ab01817b0d16b63e1c824611144f2e3
05bbcc28e71554cf13d1cc76c05329af1eb42a2966a8ec4738ef0310b9cbc555.
```

These executables are unchanged from the accepted Windows 0.6.1 package; runtime-version metadata and plugin release descriptors were updated to 1.0.0.

The macOS package targets Apple Silicon and macOS 14.0 or newer. A clean QGIS 3.44.12-Solothurn profile test installed both runtimes, prepared an AVAC case, and completed the simulation with the expected time-dependent and peak outputs. This test was performed on the available Apple Silicon system, not on a separate macOS 14 installation.

Linux is also supported by an installable package. Platform availability and installation tests do not establish numerical equivalence across operating systems. A common scientific test suite with stated tolerances is needed for that comparison.

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